

Assignment 4

Phusaeng Natthaphat (04B23054)

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1 section

Langevin 方程式を解析的に解いてみる。

$$\begin{aligned} m \frac{d}{dt} \mathbf{v}(t) &= -\zeta \mathbf{v}(t) + \mathbf{F}_B(t) \\ \langle \mathbf{F}_B(t) \cdot \mathbf{F}_B(t') \rangle &= 2d\zeta k_B T \delta(t - t') \end{aligned} \quad (1)$$

d はベクトルの次元である。ただし、 $\langle \dots \rangle$ は何らかの平均を取る動作とする。 $\mathbf{v}(t)$ の一般解は式 17 から次である。

$$\mathbf{v}(t) = e^{-\frac{\zeta}{m}t} \mathbf{v}(0) + \frac{1}{m} \int_0^t e^{-\frac{\zeta}{m}(t-t')} \mathbf{F}_B(t') dt' \quad (2)$$

The first term contains initial information of the particle, while the second term describes the overall effect of randomness on particle's dynamic. After long enough time t passes, the first term decays exponentially, leaving only the randomness contribution. I will write this physical intuition of long term (or steady state) behavior as $t \rightarrow \infty$. (Notice that I only take $t \rightarrow \infty$ in the boundary of integral not inside. This is purely physical intuition that I ignore decay in the first term and capture the cumulative effect of random noise)

$$\mathbf{v}(t) \sim \frac{1}{m} \int_0^\infty e^{-\frac{\zeta}{m}(t-t')} \mathbf{F}_B(t') dt' \quad (3)$$

In the following analysis we restrict ourselves to only investigate the long term behaviour of particle. I will modify (3) and will repeatedly use (4). Change of variable $u = t - t'$ and take the boundary of integral as $t \rightarrow \infty$. A lot of fishy mathematical manipulation happen, I'm not going to verify any of these but I will blame everything strange on involvement of ∞ .

$$\begin{aligned} \mathbf{v}(t) &\sim \frac{1}{m} \int_t^{t-\infty} e^{-\frac{\zeta}{m}(u)} \mathbf{F}_B(t-u) (-du) \\ &= \frac{1}{m} \int_0^\infty e^{-\frac{\zeta}{m}(u)} \mathbf{F}_B(t-u) du \end{aligned} \quad (4)$$

One significant remark to be made is, the information about initial conditions of the Langevin equation is lost under long time (steady state) behavior. This initial conditions-independence will play a crucial role in the simulation, because we don't have to specify the initial conditions of each particle, we can generate an ensemble of particles by dividing one long string of i.i.d. random process into several sections.

Here I took some liberty to define Auto Correlation Function $C(t_1, t_2)$.

$$C(t_1, t_2) = \langle \mathbf{v}(t_1) \cdot \mathbf{v}(t_2) \rangle \quad (5)$$

$C(t_1, t_2)$ の解析解を求めよう。I will follow the proof in Zwanzig, Nonequilibrium Statistical Mechanics; however, some mathematical/physical remarks will be made and discussed along the derivation.

First, substitute (4) into (5) and take the average $\langle \dots \rangle$.

$$\begin{aligned} \langle \mathbf{v}(t_1) \cdot \mathbf{v}(t_2) \rangle &= \frac{1}{m^2} \int_0^\infty \int_0^\infty e^{-\frac{\zeta}{m}(u+s)} \langle \mathbf{F}_B(t_1 - u) \cdot \mathbf{F}_B(t_2 - s) \rangle ds du \\ &= \frac{1}{m^2} \int_0^\infty \int_0^\infty e^{-\frac{\zeta}{m}(u+s)} 2d\zeta k_B T \delta(t_1 - u - t_2 + s) ds du \\ &= \frac{2d\zeta k_B T}{m^2} \int_0^\infty e^{-\frac{\zeta}{m}(2u - t_1 + t_2)} du \\ &= \frac{dk_B T}{m} e^{-\frac{\zeta}{m}(t_2 - t_1)} \end{aligned}$$

Or

$$\langle \mathbf{v}(t_1) \cdot \mathbf{v}(t_2) \rangle = \frac{dk_B T}{m} e^{-\frac{\zeta}{m}(t_1 - t_2)} \quad (6)$$

if you chose to substitute u first. In summary,

$$C(t_1, t_2) = \langle \mathbf{v}(t_1) \cdot \mathbf{v}(t_2) \rangle = \frac{dk_B T}{m} e^{-\frac{\zeta}{m}|t_1 - t_2|} \quad (7)$$

This peculiar property that $C(t_1, t_2)$ only depends on time difference $|t_1 - t_2|$ arises from nature of δ -function.

Next, let's consider Mean Square displacement. $\Delta \mathbf{r}(t)$ は次のように定義する。

$$\Delta \mathbf{r}(t) := \mathbf{r}(t) - \mathbf{r}(0) = \int_0^t \mathbf{v}(t') dt' \quad (8)$$

Mean Square Displacement (MSD) is defined as

$$\begin{aligned} \langle \Delta \mathbf{r}(t)^2 \rangle &= \int_0^t \int_0^t \langle \mathbf{v}(t_1) \cdot \mathbf{v}(t_2) \rangle dt_1 dt_2 \\ &= \int_0^t \int_0^t \frac{dk_B T}{m} e^{-\frac{\zeta}{m}|t_1 - t_2|} dt_1 dt_2 \end{aligned}$$

This integral can be split into 2 symmetric parts

$$\begin{aligned}
&= 2 \int_0^t \int_0^{t_2} \frac{dk_B T}{m} e^{-\frac{\zeta}{m}(t_2-t_1)} dt_1 dt_2 \\
&= \frac{2dk_B T}{m} \int_0^t e^{-\frac{\zeta}{m}t_2} \frac{m}{\zeta} \left\{ e^{\frac{\zeta}{m}t_2} - 1 \right\} dt_2 \\
&= \frac{2dk_B T}{\zeta} \left\{ t + \frac{m}{\zeta} \left[e^{-\frac{\zeta}{m}t} - 1 \right] \right\}
\end{aligned}$$

最終的に

$$\langle \Delta \mathbf{r}(t)^2 \rangle = \frac{2dk_B T}{\zeta} \left\{ t + \frac{m}{\zeta} e^{-\zeta t/m} - \frac{m}{\zeta} \right\} \quad (9)$$

と証明できた。これを無次元化する。

$$\begin{aligned}
\Delta \mathbf{r} &= a \Delta \tilde{\mathbf{r}} \\
t &= t_0 \tilde{t} = \frac{m}{\zeta} \tilde{t}
\end{aligned} \quad (10)$$

式 9は次の形をする。

$$\begin{aligned}
\langle \Delta \tilde{\mathbf{r}}(\tilde{t})^2 \rangle &= 2d \frac{k_B T m}{a^2 \zeta^2} \left\{ \tilde{t} + e^{-\tilde{t}} - 1 \right\} \\
&:= 2dT^* \left\{ \tilde{t} + e^{-\tilde{t}} - 1 \right\}
\end{aligned} \quad (11)$$

式 7は次である。

$$\begin{aligned}
\tilde{C}(\tilde{t}_1, \tilde{t}_2) &= \frac{t_0^2}{a^2} \frac{dk_B T}{m} e^{-\frac{\zeta}{m} t_0 \tilde{t}} \\
&:= dT^* e^{-\tilde{t}}
\end{aligned} \quad (12)$$

2 結果

In the following simulation, everything is done in $d = 3$ dimensions.

2.1 Normal Random Number generator

Using Masaglia Polar method one can generate random number that follows Gaussian distribution. In this section, I only verify that is indeed the case.

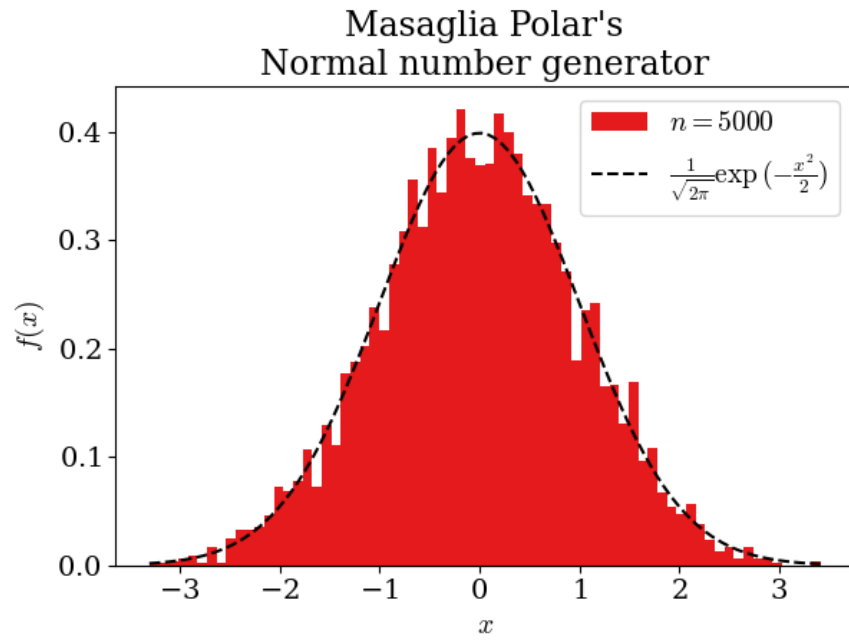


図 1: Masaglia Polar method

2.2 Simulation Result

Mean Square Displacement (MSD) の時間変化は以下である (図 2)。

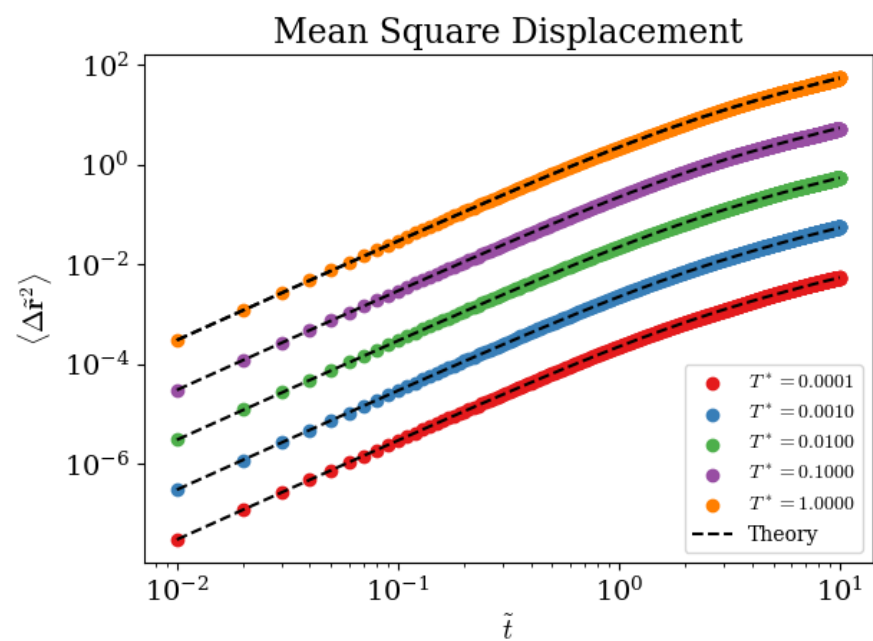


図 2: Mean Square Displacement

また、Auto Correlation Function のグラフは以下である (図 3)。

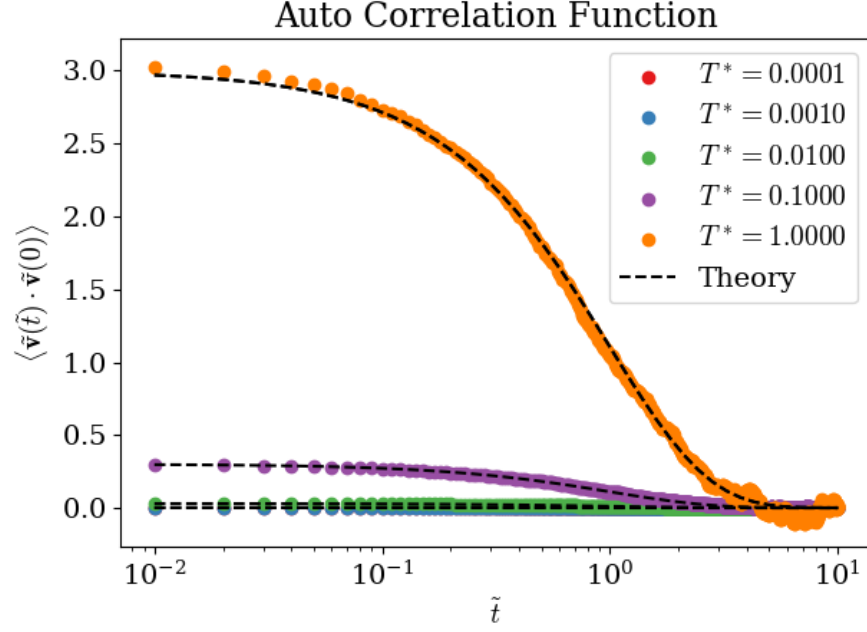


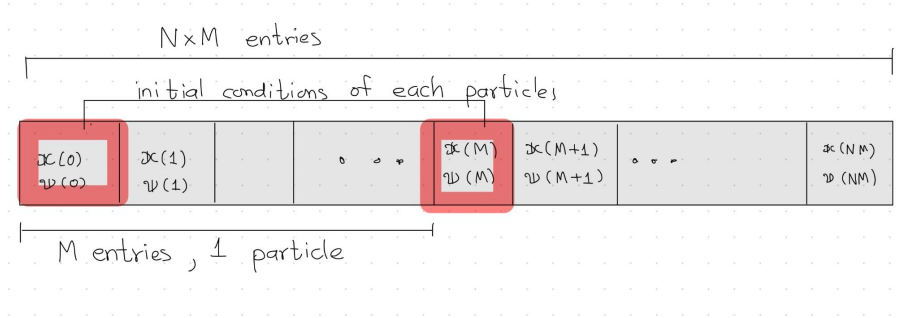
図 3: Mean Square Displacement

図からわかるように MSD のシミュレーション結果が理論値とよく合っている。一方、Auto Correlation Function のシミュレーション結果は時間が経つにつれて、理論値から離れていく。

3 Discussion

Let the number of timestep $M = 1000$ and number of particle of ensemble be $N = 1000$. In the simulation, I set the initial conditions only once in the beginning of the loop. After that, `langevin.cpp` runs the discretized Langevin Equation of Motion for another $N \times M$ times. What we get is a $N \times M$ entries of position/velocity vectors.

Then we dividie this $N \times M$ long sequence into N blocks, each with length M . Each block represent the dynamic of 1 particle over $M \times \Delta t$ time period. Each blocks all have different initial conditions, we already saw a glimpse of initial conditions-independence when we restricted ourselves to study the steady state behavior of particles.



⊠ 4: Ensemble of Particle

I think we can say,

1. We can divide random sequence $\{x_i\}_{i=1}^{NM}$ governed by Langevin equation into i.i.d. subsequences $\{x_{i_k}\}_{k=1}^M$ because the Langevin process itself is i.i.d.
2. Since I specified the initial conditions only once in the beginning of sequence $\{x_i\}$, one can say that $\exists N_0$ s.t. $\forall i \geq N_0$ $\{x_i\}$ is a steady states which (3) holds.
3. Despite not knowing the exact value of N_0 but with enough iteration, most of the subsequences $\{x_{i_k}\}$ can said to be a steady state. Hence, by doing ensemble average we can verify validity of (12) and (11).

To test my explanation `langevin_test.cpp` specifies the initial conditions of every particle in the iteration. The result is below. Clearly, at first the simulation deviate from Theory this shows the non-steady state behavior. However, after letting the iteration goes up, simulation and theory finally converge, this shows that the steady state behaviors finally appear after long enough time $N_0 \times \Delta t$.

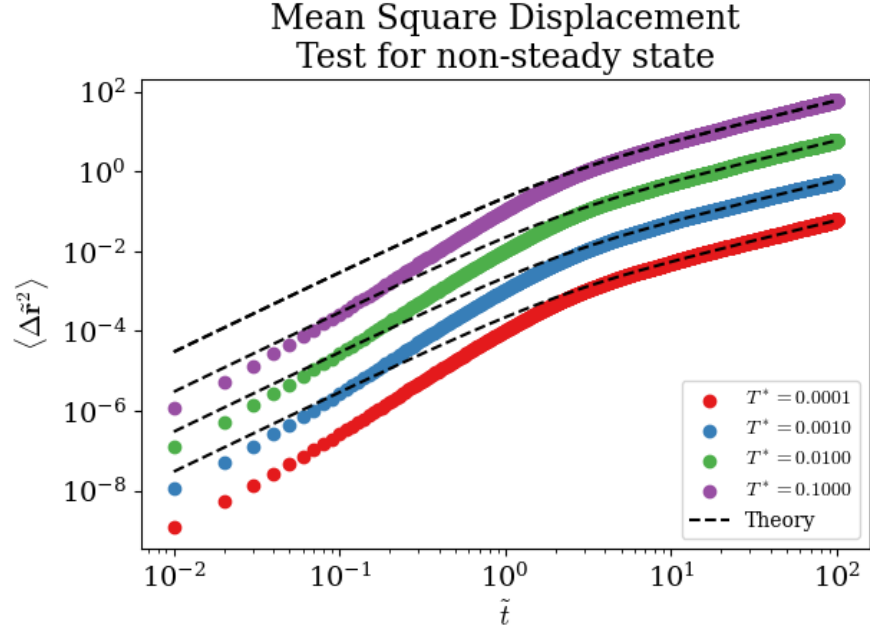


図 5: MSD with initial conditions

4 Appendix

$$\dot{x}(t) = ax(t) + b(t) \quad (13)$$

の一般解を求めよう。まず

$$x(t) = e^{at}y(t) \quad (14)$$

とおく。式 13が

$$\dot{y}(t) = e^{-at}b(t) \quad (15)$$

となり、

$$y(t) = y(0) + \int_0^t e^{-at'} b(t') dt' \quad (16)$$

だと分かる。すなわち、式 13の一般解が

$$x(t) = e^{at}x(0) + \int_0^t e^{a(t-t')} b(t') dt' \quad (17)$$

である。